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(54) **FOLDABLE ELECTRONIC DEVICE
INCLUDING HINGE LOCKING STRUCTURE**

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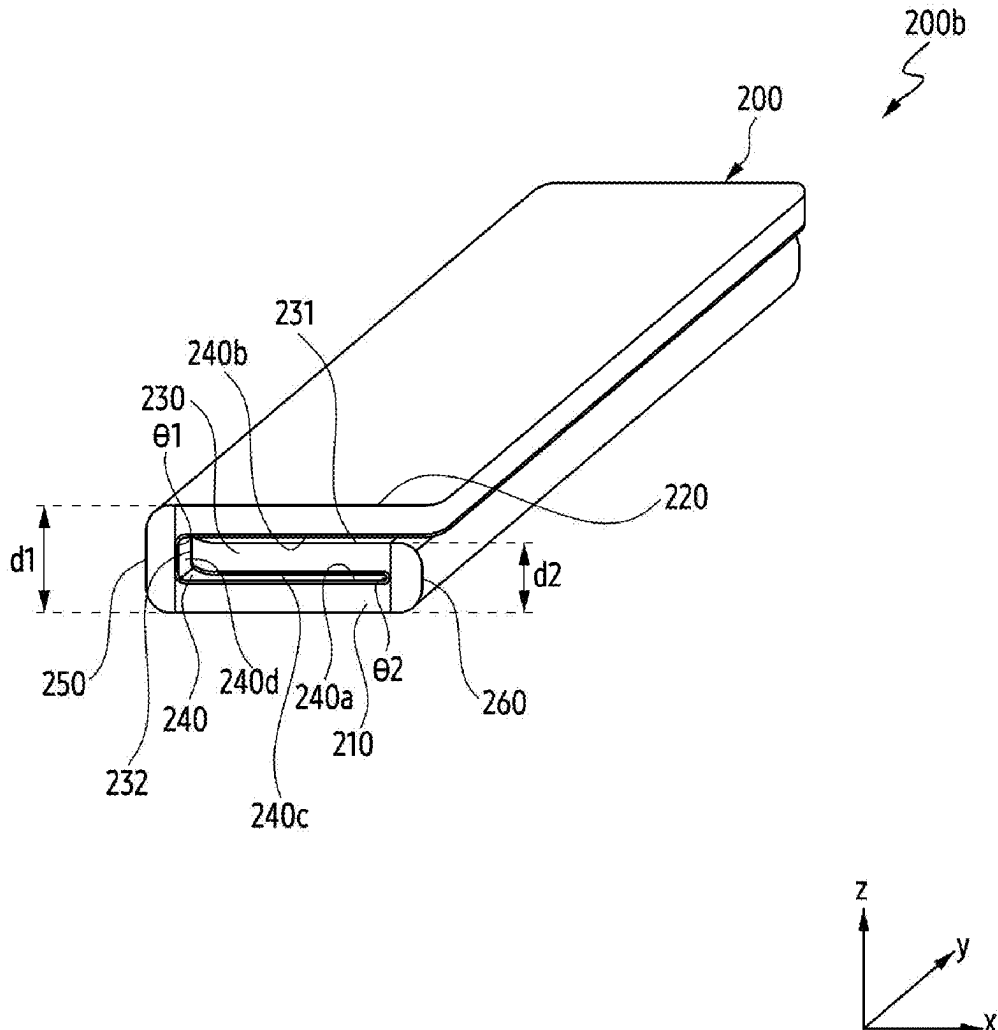
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(57) **ABSTRACT**

An example electronic device may include a first housing including a first surface, a second housing including a second surface, a third housing including a third surface, a first hinge structure rotatably connecting the second housing and the first housing, a second hinge structure rotatably connecting the first housing and the third housing, a flexible display disposed on the first surface, the second surface and the third surface, a first magnet disposed in the second housing, and a hinge locking structure which, in an abnormal state of the electronic device in which a portion of the first surface is adjacent to the second surface, faces the first magnet and which includes a pin insertable into a groove formed in at least one shaft in the second hinge structure.



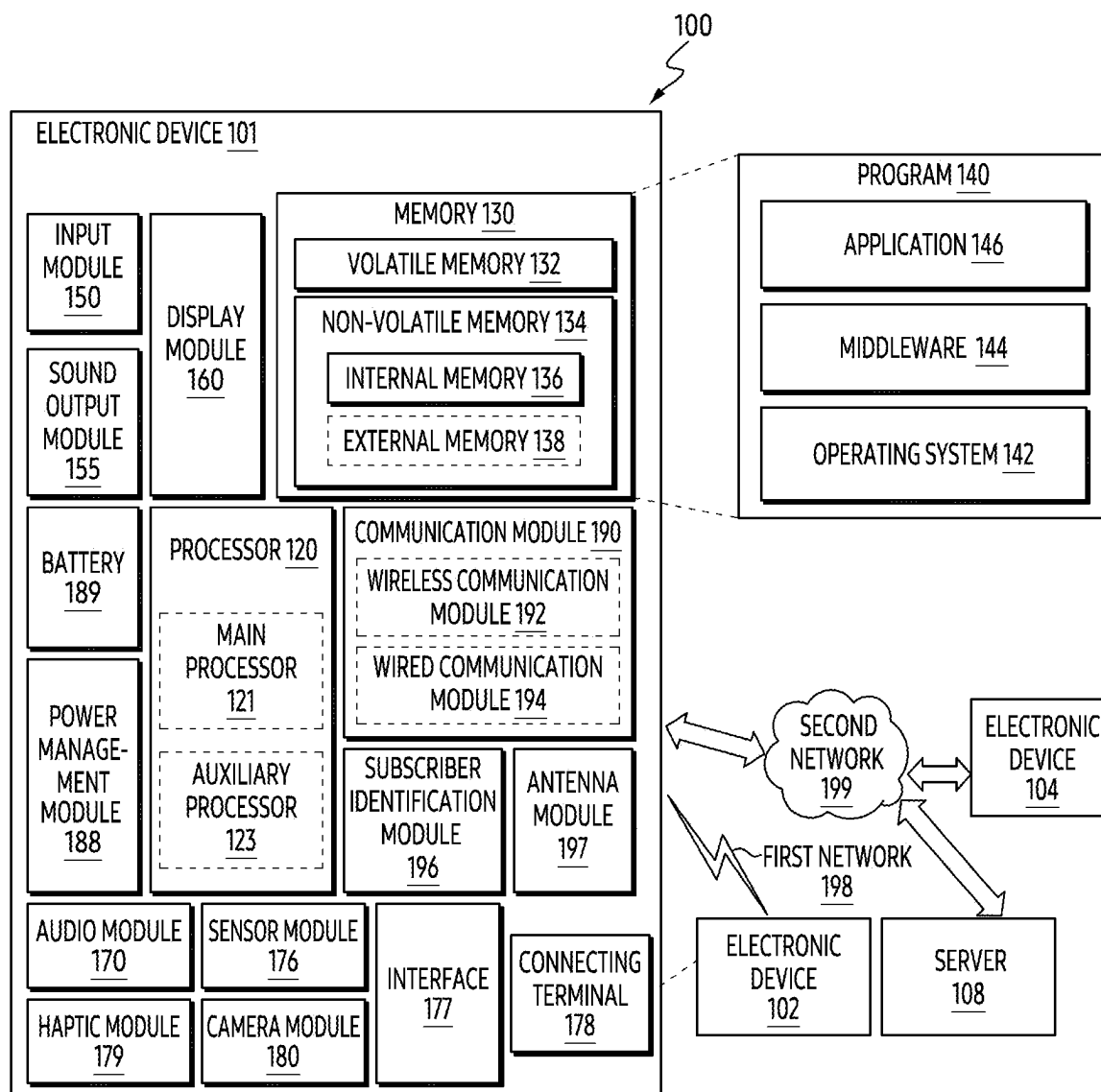


FIG. 1

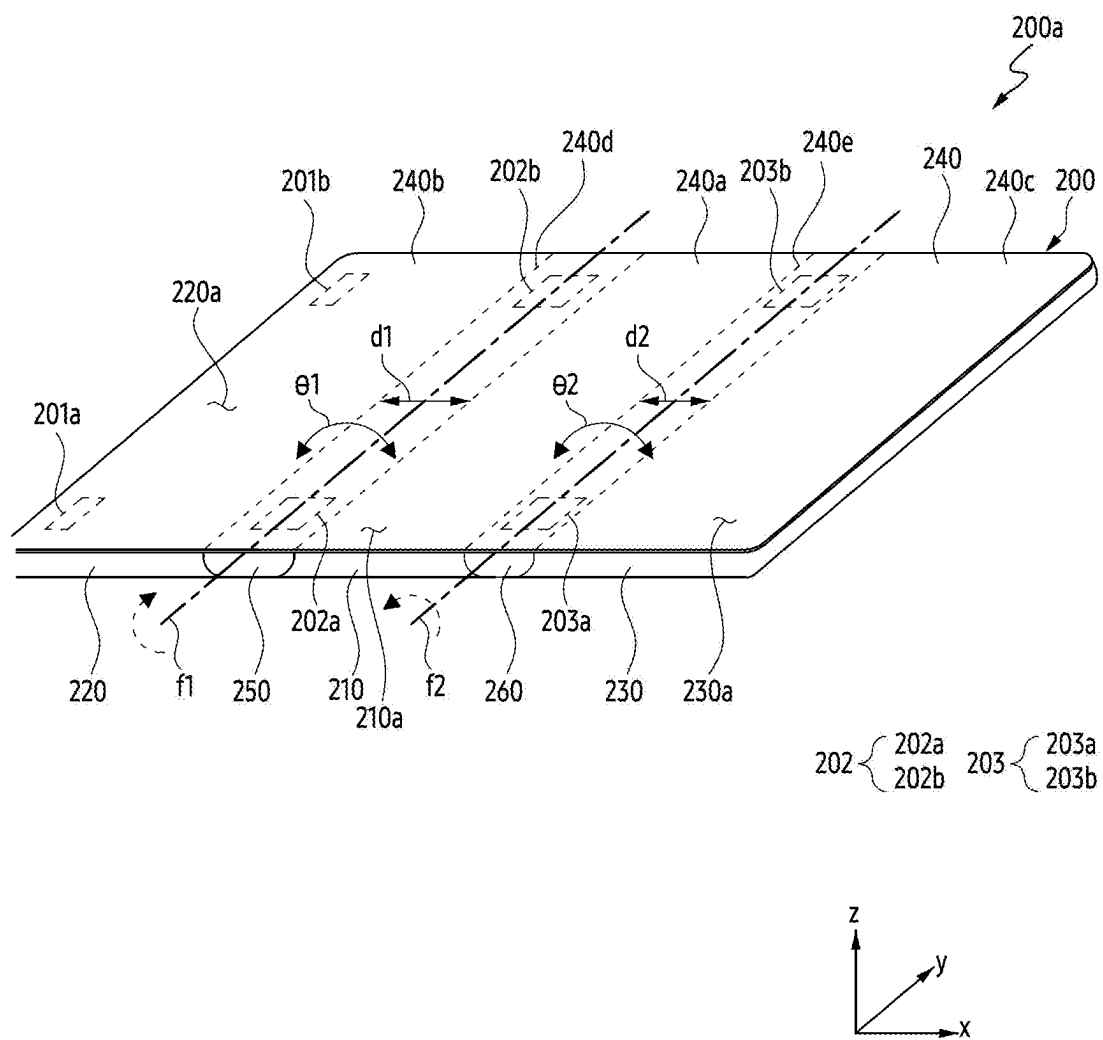


FIG. 2A

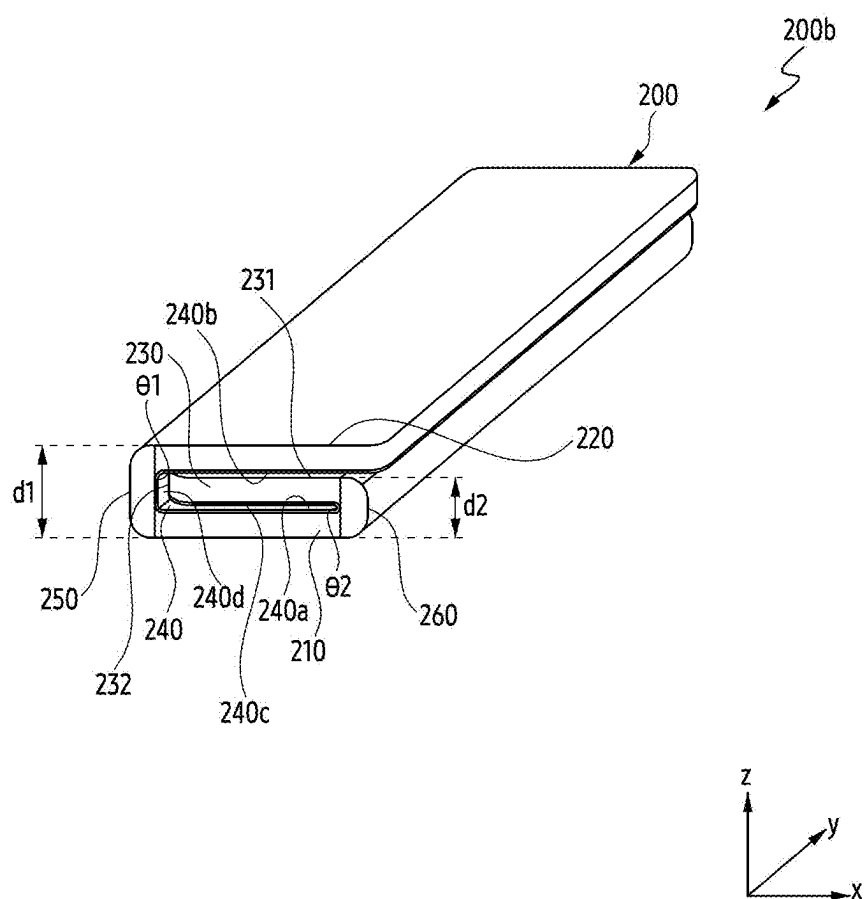


FIG. 2B

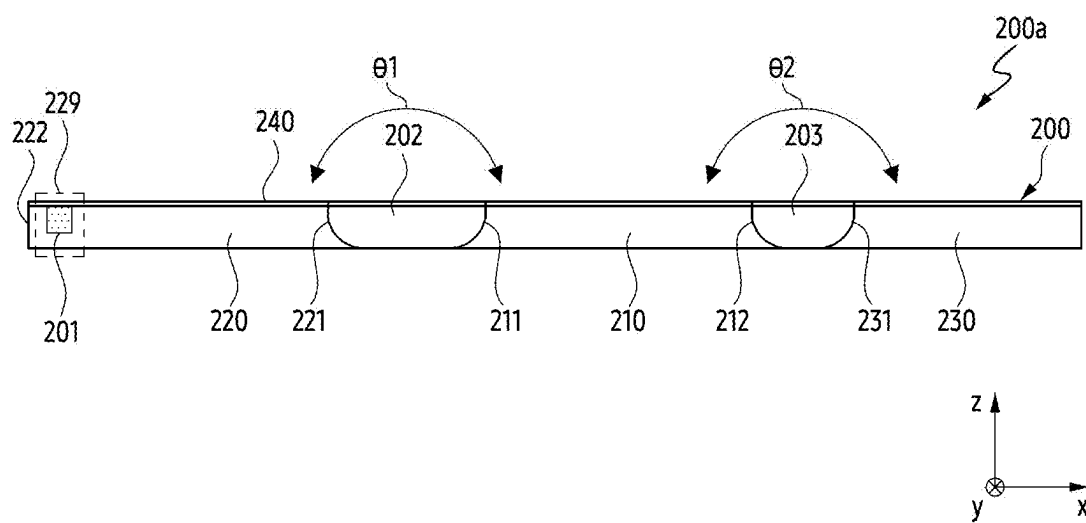


FIG. 3A

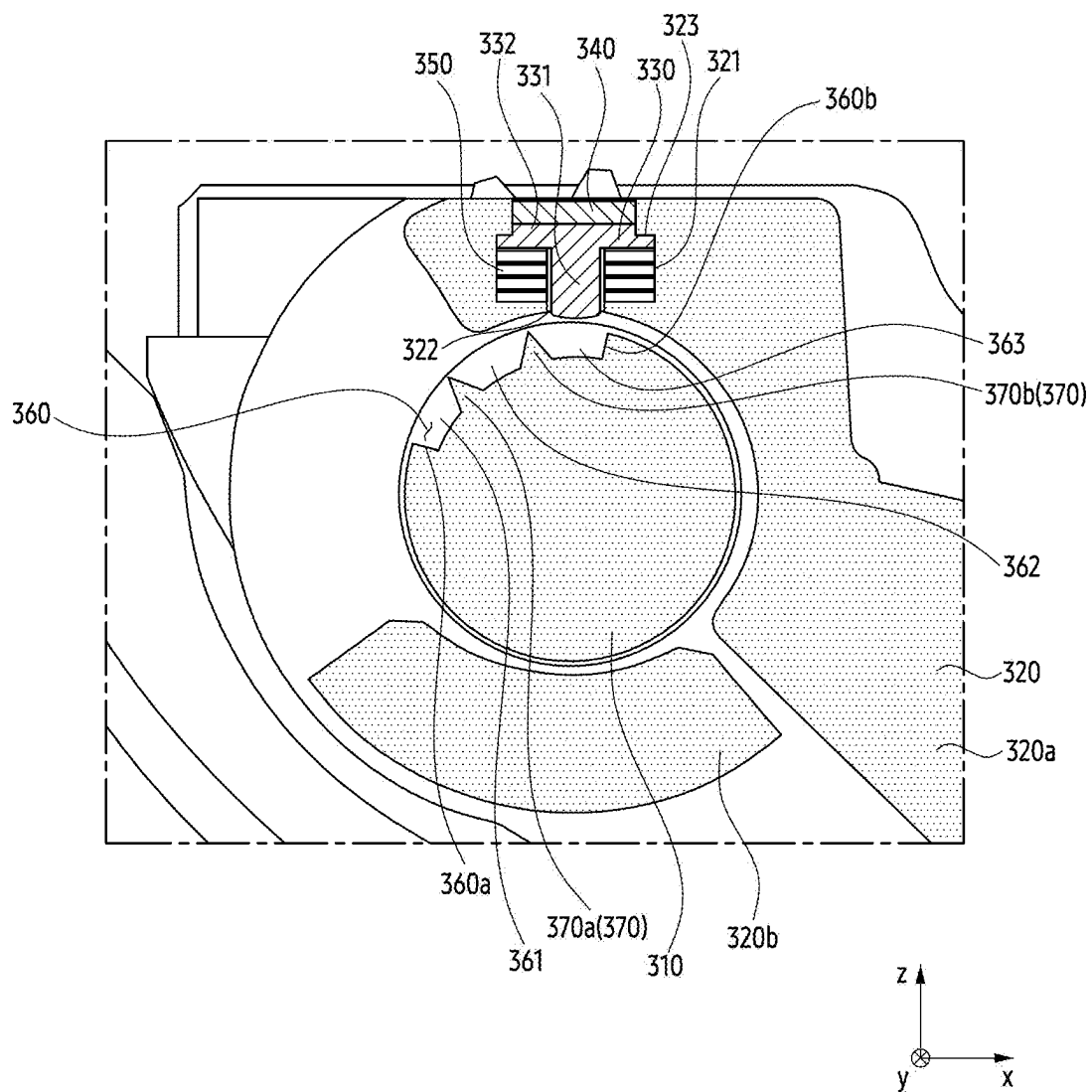


FIG. 3B

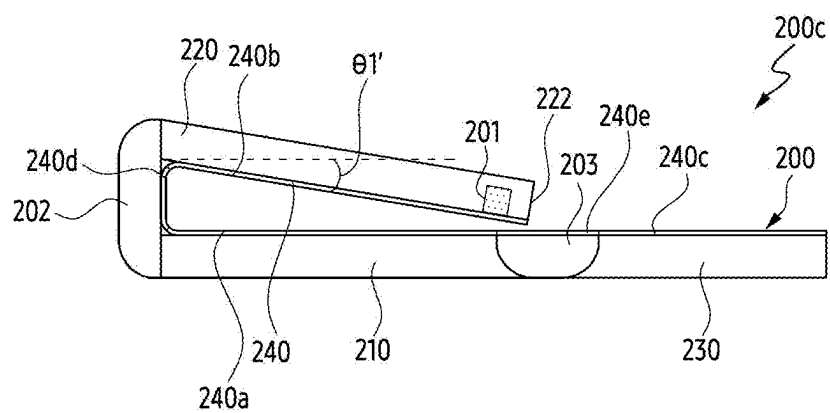


FIG. 4A

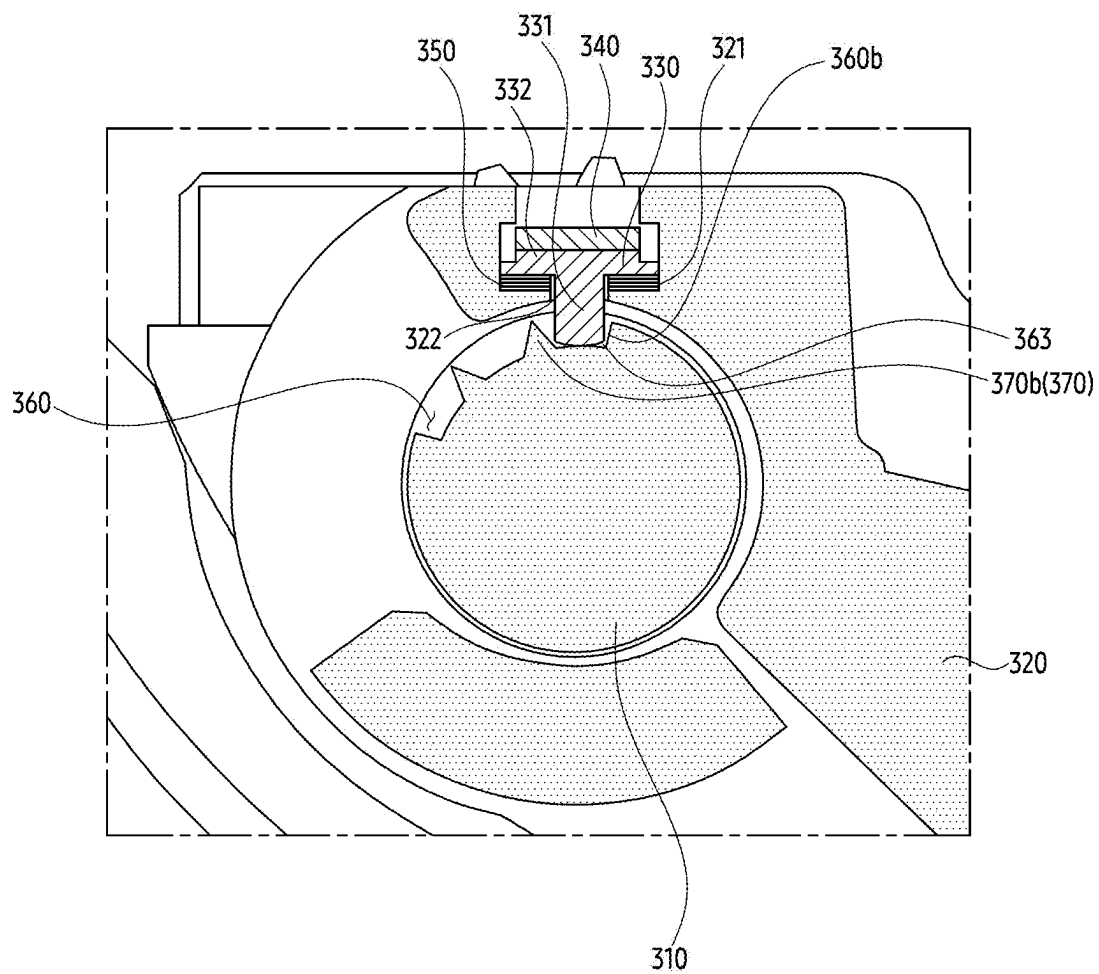


FIG. 4B

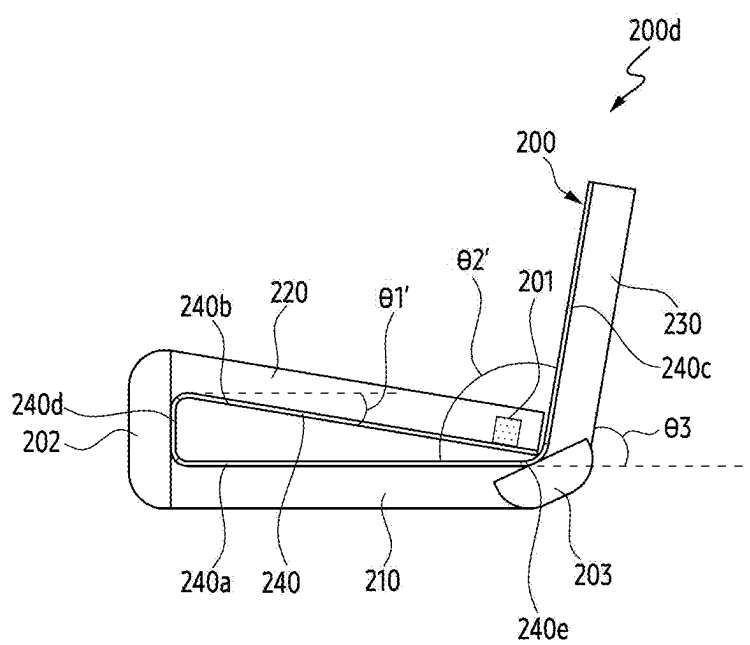


FIG. 5A

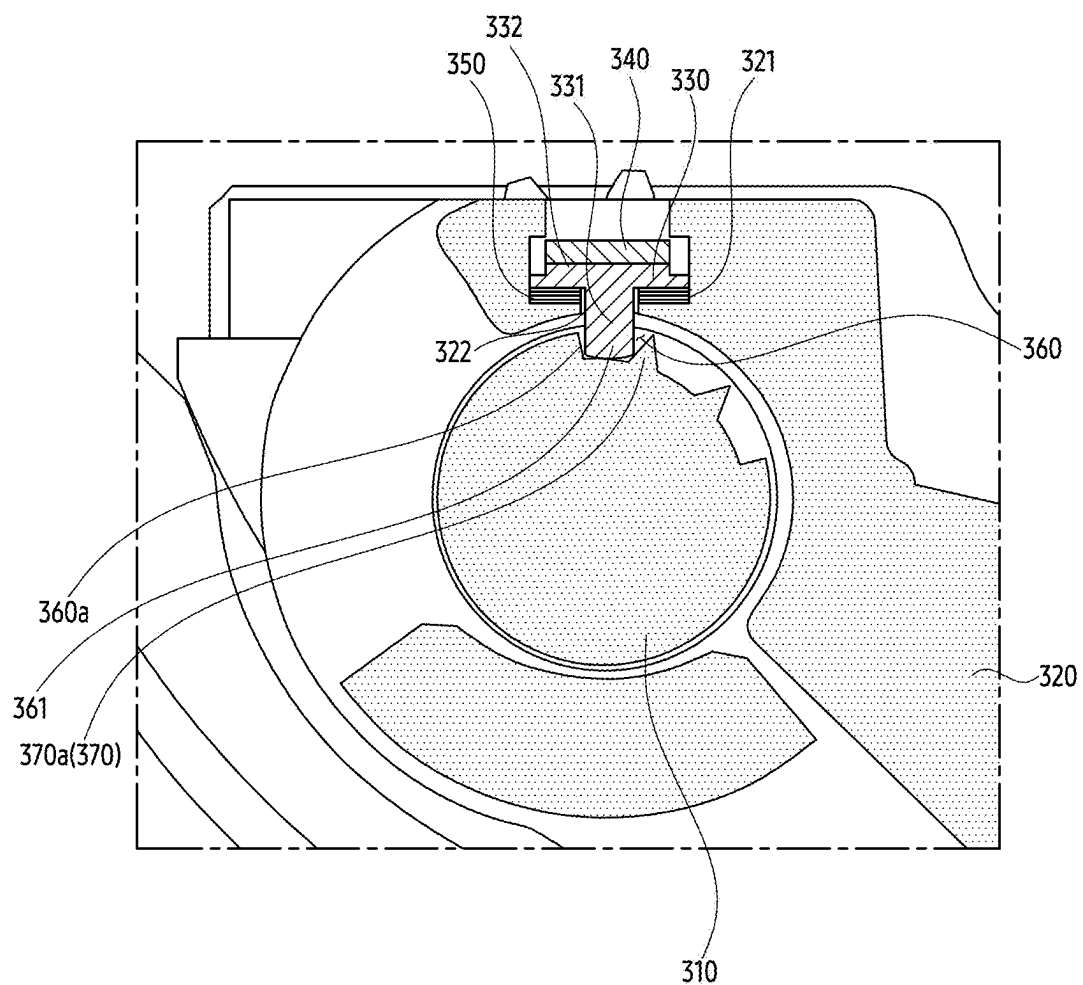


FIG. 5B

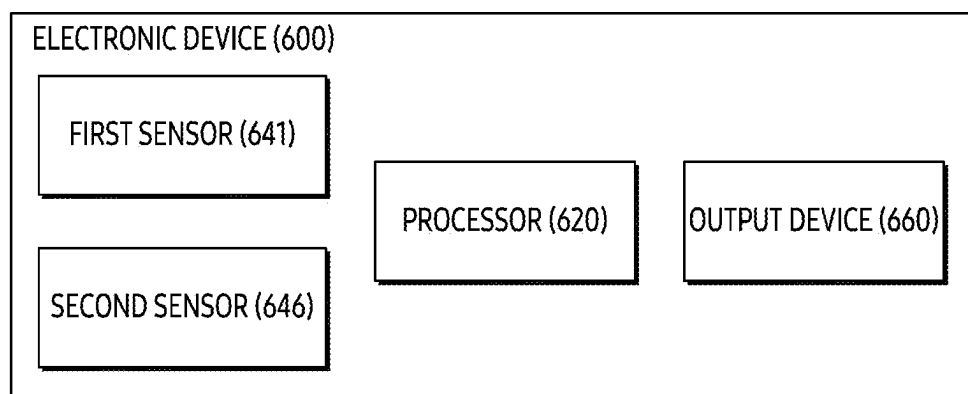


FIG. 6

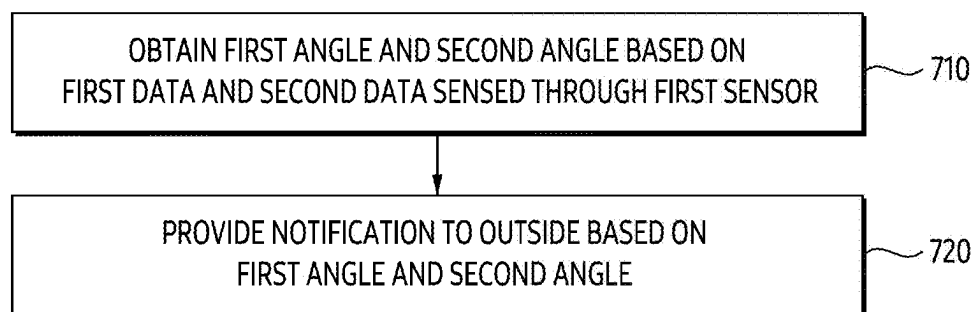


FIG. 7

FOLDABLE ELECTRONIC DEVICE INCLUDING HINGE LOCKING STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/190,632, filed Mar. 27, 2023, which is a continuation of International Application No. PCT/KR2023/003718, designating the United States, filed on Mar. 21, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0078823, filed on Jun. 28, 2022 in the Korean Intellectual Property Office and to Korean Patent Application No. 10-2022-0098134, filed on Aug. 5, 2022 in the Korean Intellectual Property Office. The disclosures of each of these applications are incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] The disclosure relates to a foldable electronic device including a hinge locking structure.

Description of Related Art

[0003] An electronic device including a large screen display may increase user utilization. As demand for electronic devices having high portability increases, the electronic devices may be provided with a deformable display. The deformable display may be deformable in a slidable manner, foldable manner, or rollable manner.

SUMMARY

[0004] A deformable display may have structurally weak rigidity. The deformable display may be damaged during use or by use contrary to an intended usage. An electronic device described herein may provide a structure and a method for preventing damage to the deformable display.

[0005] The technical problems to be achieved in this disclosure are not limited to those described above, and other technical problems not mentioned herein will be clearly understood from the following description.

[0006] According to an example embodiment, an electronic device may include a first housing including a first surface, a second housing including a second surface, and a third housing including a third surface. The electronic device may further include a first hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the second housing facing a first side surface of the first housing. The electronic device may further include a second hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the third housing facing a second side surface of the first housing opposite to the first side surface. The electronic device may further include a flexible display disposed on the first surface, the second surface and the third surface. The electronic device may include a first magnet disposed in the second housing, and a hinge locking structure (including, e.g., a hinge lock) which, in an abnormal state of the electronic device in which a portion of the first surface is adjacent to the second surface, faces the first magnet and which includes a pin insertable into a groove formed in at least one shaft in the second hinge structure. The first hinge

structure and the second hinge structure may be configured to provide an unfolded state in which the first surface, the second surface, and the third surface face the same direction or a folded state in which at least one of the first surface, the second surface, or the third surface faces different directions. The hinge locking structure may be configured such that the pin is inserted into the groove by a force of the first magnet when the electronic device is in the abnormal state, and the pin is disengaged from the groove as the electronic device changes from the abnormal state to the unfolded state.

[0007] According to an embodiment, an electronic device may include a first housing including a first surface, a second housing including a second surface, and a third housing including a third surface. The electronic device may further include a first hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the second housing facing a first side surface of the first housing. The electronic device may further include a second hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the third housing facing a second side surface of the first housing opposite to the first side surface. The electronic device may further include a flexible display disposed on the first surface, the second surface and the third surface. The electronic device may further include a first sensor disposed in the second housing and configured to obtain first data related to a posture of the second housing, and a second sensor disposed in the third housing and configured to obtain second data related to a posture of the third housing. The electronic device may include a first magnet disposed in the second housing, and a hinge locking structure (including, e.g., a hinge lock) which, in an abnormal state of the electronic device in which a portion of the first surface is adjacent to the second surface, faces the first magnet and includes a pin insertable into a groove formed in at least one shaft in the second hinge structure. The first hinge structure and the second hinge structure may be configured to provide an unfolded state in which the first surface, the second surface, and the third surface face the same direction or a folded state in which at least one of the first surface, the second surface, or the third surface face different directions. An overlapped area of the flexible display and the first hinge structure may be wider than an overlapped area of the flexible display and the second hinge structure, in the unfolded state when the flexible display is viewed from above.

[0008] According to an embodiment, a foldable electronic device including a hinge locking structure can reduce damage to a display or the electronic device due to folding that is contrary to intended usage.

[0009] The effects that can be obtained from the present disclosure are not limited to those described above, and any other effects not mentioned herein will be clearly understood from the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and/or other aspects of the disclosure will be more apparent by describing certain embodiments of the disclosure with reference to the accompanying drawings, in which:

[0011] FIG. 1 is a block diagram of an example electronic device in a network environment according to various embodiments;

[0012] FIG. 2A illustrates an example of an unfolded state of an example electronic device according to various embodiments;

[0013] FIG. 2B illustrates an example of a fully folded state of an example electronic device according to various embodiments;

[0014] FIG. 3A illustrates an example of arrangement of a hinge locking structure in an unfolded state of an example electronic device according to various embodiments;

[0015] FIG. 3B illustrates an example hinge locking structure in an unfolded state of an example electronic device according to various embodiments;

[0016] FIG. 4A illustrates an example of arrangement of a hinge locking structure in a third state of an example electronic device according to various embodiments;

[0017] FIG. 4B illustrates an example hinge locking structure in a third state of an example electronic device according to various embodiments;

[0018] FIG. 5A illustrates an example of arrangement of a hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments;

[0019] FIG. 5B illustrates an example hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments;

[0020] FIG. 6 is a block diagram of an example electronic device according to various embodiments; and

[0021] FIG. 7 is a flowchart illustrating an example operation of detecting an abnormal state of an example electronic device according to various embodiments.

DETAILED DESCRIPTION

[0022] FIG. 1 is a block diagram illustrating an example electronic device 101 in a network environment 100 according to various embodiments.

[0023] Referring to FIG. 1, the electronic device 101 in the network environment 100 may communicate with an electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or at least one of an electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment, the electronic device 101 may communicate with the electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, memory 130, an input module 150, a sound output module 155, a display module 160, an audio module 170, a sensor module 176, an interface 177, a connecting terminal 178, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In various embodiments, at least one of the components (e.g., the connecting terminal 178) may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments, some of the components (e.g., the sensor module 176, the camera module 180, or the antenna module 197) may be implemented as a single component (e.g., the display module 160).

[0024] The processor 120 may execute, for example, software (e.g., a program 140) to control at least one other component (e.g., a hardware or software component) of the electronic device 101 coupled with the processor 120, and

may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor 120 may store a command or data received from another component (e.g., the sensor module 176 or the communication module 190) in volatile memory 132, process the command or the data stored in the volatile memory 132, and store resulting data in non-volatile memory 134. According to an embodiment, the processor 120 may include a main processor 121 (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor 123 (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor 121. For example, when the electronic device 101 includes the main processor 121 and the auxiliary processor 123, the auxiliary processor 123 may be adapted to consume less power than the main processor 121, or to be specific to a specified function. The auxiliary processor 123 may be implemented as separate from, or as part of the main processor 121.

[0025] The auxiliary processor 123 may control at least some of functions or states related to at least one component (e.g., the display module 160, the sensor module 176, or the communication module 190) among the components of the electronic device 101, instead of the main processor 121 while the main processor 121 is in an inactive (e.g., sleep) state, or together with the main processor 121 while the main processor 121 is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor 123 (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module 180 or the communication module 190) functionally related to the auxiliary processor 123. According to an embodiment, the auxiliary processor 123 (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device 101 where the artificial intelligence is performed or via a separate server (e.g., the server 108). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0026] The memory 130 may store various data used by at least one component (e.g., the processor 120 or the sensor module 176) of the electronic device 101. The various data may include, for example, software (e.g., the program 140) and input data or output data for a command related thereto. The memory 130 may include the volatile memory 132 or the non-volatile memory 134.

[0027] The program 140 may be stored in the memory 130 as software, and may include, for example, an operating system (OS) 142, middleware 144, or an application 146.

[0028] The input module 150 may receive a command or data to be used by another component (e.g., the processor 120) of the electronic device 101, from the outside (e.g., a user) of the electronic device 101. The input module 150 may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0029] The sound output module 155 may output sound signals to the outside of the electronic device 101. The sound output module 155 may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0030] The display module 160 may visually provide information to the outside (e.g., a user) of the electronic device 101. The display module 160 may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module 160 may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0031] The audio module 170 may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module 170 may obtain the sound via the input module 150, or output the sound via the sound output module 155 or a headphone of an external electronic device (e.g., an electronic device 102) directly (e.g., wiredly) or wirelessly coupled with the electronic device 101.

[0032] The sensor module 176 may detect an operational state (e.g., power or temperature) of the electronic device 101 or an environmental state (e.g., a state of a user) external to the electronic device 101, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module 176 may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0033] The interface 177 may support one or more specified protocols to be used for the electronic device 101 to be coupled with the external electronic device (e.g., the electronic device 102) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface 177 may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0034] A connecting terminal 178 may include a connector via which the electronic device 101 may be physically connected with the external electronic device (e.g., the electronic device 102). According to an embodiment, the connecting terminal 178 may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0035] The haptic module 179 may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized

by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module 179 may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0036] The camera module 180 may capture a still image or moving images. According to an embodiment, the camera module 180 may include one or more lenses, image sensors, image signal processors, or flashes.

[0037] The power management module 188 may manage power supplied to the electronic device 101. According to an embodiment, the power management module 188 may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0038] The battery 189 may supply power to at least one component of the electronic device 101. According to an embodiment, the battery 189 may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0039] The communication module 190 may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device 101 and the external electronic device (e.g., the electronic device 102, the electronic device 104, or the server 108) and performing communication via the established communication channel. The communication module 190 may include one or more communication processors that are operable independently from the processor 120 (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module 190 may include a wireless communication module 192 (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module 194 (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network 198 (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network 199 (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module 192 may identify and authenticate the electronic device 101 in a communication network, such as the first network 198 or the second network 199, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module 196.

[0040] The wireless communication module 192 may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module 192 may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module 192 may support

various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module 192 may support various requirements specified in the electronic device 101, an external electronic device (e.g., the electronic device 104), or a network system (e.g., the second network 199). According to an embodiment, the wireless communication module 192 may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0041] The antenna module 197 may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device 101. According to an embodiment, the antenna module 197 may include an antenna including a radiating element composed of or including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module 197 may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network 198 or the second network 199, may be selected, for example, by the communication module 190 (e.g., the wireless communication module 192) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module 190 and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module 197.

[0042] According to various embodiments, the antenna module 197 may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0043] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0044] According to an embodiment, commands or data may be transmitted or received between the electronic device 101 and the external electronic device 104 via the server 108 coupled with the second network 199. Each of the electronic devices 102 or 104 may be a device of a same type as, or a different type, from the electronic device 101. According to an embodiment, all or some of operations to be executed at the electronic device 101 may be executed at one or more of the external electronic devices 102, 104, or 108. For example, if the electronic device 101 should perform a

function or a service automatically, or in response to a request from a user or another device, the electronic device 101, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device 101. The electronic device 101 may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device 101 may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device 104 may include an internet-of-things (IoT) device. The server 108 may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device 104 or the server 108 may be included in the second network 199. The electronic device 101 may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0045] FIG. 2A illustrates an example of an unfolded state of an example electronic device according to various embodiments. FIG. 2B illustrates an example of a fully folded state of an example electronic device according to various embodiments.

[0046] A display (e.g., the display module 160 of FIG. 1) of an electronic device (e.g., the electronic device 101 of FIG. 1) may be folded at least once. The display may be referred to, for example, as a foldable display or a flexible display in terms of having a foldable structure. The electronic device may be referred to, for example, as a foldable electronic device or a flexible electronic device in terms of including a foldable display.

[0047] Referring to FIG. 2A, an electronic device 200 may be unfolded in a first state 200a. The electronic device 200 may include a first housing 210, a second housing 220, and a third housing 230. The first housing 210 may include a first surface 210a. The second housing 220 may include a second surface 220a. The third housing 230 may include a third surface 230a. The first state 200a may be a state in which the first surface 210a, the second surface 220a, and the third surface 230a face substantially the same first direction (e.g., the z-axis direction). The first state 200a may be referred to, for example, as an unfolded state in which the first housing 210, the second housing 220, and the third housing 230 are unfolded. In the first state 200a, the first housing 210, the second housing 220, and the third housing 230 may be substantially planar. For example, since the first housing 210, the second housing 220, and the third housing 230 include a substantially planar surface, the first surface 210a, the second surface 220a, and the third surface 230a may be planar.

[0048] The electronic device 200 may include a first hinge structure 202 (202a and 202b) and a second hinge structure 203 (203a and 203b). The first hinge structure 202 may include a plurality of first hinge structures 202a and 202b. The second hinge structure 203 may include a plurality of second hinge structures 203a and 203b. The first hinge

structures **202a** and **202b** may be disposed between the first housing **210** and the second housing **220**. The first hinge structures **202a** and **202b** may rotatably connect the second housing **220** to the first housing **210**. For example, the second housing **220** may rotate with respect to the first housing **210** based on a first folding shaft **f1**, through the first hinge structures **202a** and **202b**. A first angle θ_1 between the first housing **210** and the second housing **220** may be approximately 0 degrees to 180 degrees, through the rotation of the second housing **220**. The second hinge structures **203a** and **203b** may be disposed between the first housing **210** and the third housing **230**. The second hinge structures **203a** and **203b** may rotatably connect the third housing **230** to the first housing **210**. For example, the third housing **230** may rotate with respect to the first housing **210** based on a second folding shaft **f2**, through the second hinge structures **203a** and **203b**. The second angle θ_2 between the first housing **210** and the third housing **230** may be approximately 0 degrees to 180 degrees, through the rotation of the third housing **230**.

[0049] The first hinge structures **202a** and **202b** may be surrounded by the first hinge cover **250**. One (e.g., **202a**) of the first hinge structures **202a** and **202b** may be disposed at one end of the first hinge cover **250**, and another (e.g., **202b**) may be disposed at another end of the first hinge cover **250**. The second hinge structures **203a** and **203b** may be surrounded by a second hinge cover **260**. One (e.g., **203a**) of the second hinge structures **203a** and **203b** may be disposed at one end of the second hinge cover **260**, and another (e.g., **203b**) may be disposed at another end of the second hinge cover **260**.

[0050] The electronic device **200** may further include a flexible display **240**. The flexible display **240** may be disposed on one surface formed by the first housing **210**, the second housing **220**, and the third housing **230**. The flexible display **240** may be disposed on the first surface **210a**, the second surface **220a**, and the third surface **230a**. The flexible display **240** may include a first display area **240a**, a second display area **240b**, and a third display area **240c**. The first display area **240a** may be disposed on the first surface **210a** of the first housing **210**. The first display area **240a** may be an area coupled to the first surface **210a** and not deformed. The second display area **240b** may be disposed on the second surface **220a** of the second housing **220**. The second display area **240b** may be an area coupled to the second surface **220a** and not deformed. The third display area **240c** may be disposed on the third surface **230a** of the third housing **230**. The third display area **240c** may be an area coupled to the third surface **230a** and not deformed. The first display area **240a**, the second display area **240b**, and the third display area **240c** may be referred to, for example, as planar areas, in terms of forming substantially planar surface.

[0051] The flexible display **240** may further include a fourth display area **240d** and a fifth display area **240e**. The fourth display area **240d** may be disposed on the first hinge cover **250**. The fourth display area **240d** may be coupled to the first hinge cover **250** or the first hinge structures **202a** and **202b**. For example, the fourth display area **240d** may be supported by a hinge plate (not illustrated) included in the first hinge structures **202a** and **202b**. The fourth display area **240d** may be deformed as the second housing **220** rotates based on the first folding shaft **f1** with respect to the first housing **210**.

[0052] The fifth display area **240e** may be disposed on the second hinge cover **260**. The fifth display area **240e** may be coupled to the second hinge cover **260** or the second hinge structures **203a** and **203b**. For example, the fifth display area **240e** may be supported by a hinge plate (not illustrated) included in the second hinge structures **203a** and **203b**. The fifth display area **240e** may be deformed as the third housing **230** rotates based on the second folding shaft **f2** with respect to the first housing **210**.

[0053] The fourth display area **240d** may be larger than the fifth display area **240e**. For example, a first distance **d1**, which is the width of the fourth display area **240d**, may be longer (greater) than a second distance **d2**, which is the width of the fifth display area **240e**. When the electronic device **200** is converted into the folded state, the radius of curvature of the fourth display area **240d** may be greater than the radius of curvature of the fifth display area **240e**.

[0054] The electronic device **200** may further include first magnets **201a** and **201b**. The first magnets **201a** and **201b** may be disposed in the second housing **220**. For example, the first magnets **201a** may be disposed on the side surface of the second housing **220**. The first magnets **201a** may be disposed closer to another side surface facing the one side surface than the one side surface of the second housing **220** facing the first hinge cover **250**. The first magnets **201a** may at least partially overlap an inactive area disposed along the periphery of the second display area **240b** when the flexible display **240** is viewed from above. The first magnets **201a** and **201b** may be disposed to face the second hinge structures **203a** and **203b**, when the second housing **220** is folded by rotating along the first folding shaft **f1**.

[0055] Referring to FIG. 2B, the electronic device **200** may be folded in a second state **200b**. In the second state **200b**, at least one of the first surface **210a**, the second surface **220a**, or the third surface **230a** may face different directions. The third display area **240c** corresponding to the third surface **230a** may face substantially the same direction as the direction in which the second display area **240b** corresponding to the second surface **220a** is directed (e.g., the $-z$ axis direction). The third display area **240c** may face a direction distinct from a direction in which the first display area **240a** corresponding to the first surface **210a** is directed. For example, a direction in which the third display area **240c** is directed (e.g., the $-z$ axis direction) may be opposite to a direction in which the first display area **240a** is directed (e.g., the $+z$ axis direction). The third display area **240c** may face the first display area **240a**. Since the first housing **210**, the second housing **220**, and the third housing **230** are folded, the second state **200b** may be referred to, for example, as a folded state. In the second state **200b**, the first housing **210**, the second housing **220**, and the third housing **230** may face each other. For example, in the second state **200b**, the first housing **210**, the third housing **230**, and the second housing **220** may be stacked in the first direction (e.g., the z axis direction).

[0056] While changing from the first state **200a** to the second state **200b**, the second housing **220** may rotate with respect to the first housing **210** based on the first folding shaft **f1**, through the first hinge structure **202**. In the second state **200b**, the second housing **220** may face the third housing **230**. For example, in the second state **200b**, the second display area **240b** disposed on the second housing **220** may face a rear surface **231** of the third housing **230**.

[0057] Through the second hinge structure 203, the third housing 230 may rotate with respect to the first housing 210 based on the second folding shaft f2. In the second state 200b, the third housing 230 may face the first housing 210. For example, in the second state 200b, the third display area 240c disposed on the third housing 230 may face the first display area 240a disposed on the first housing 210. The side surface 232 of the third housing 230 may face the fourth display area 240d.

[0058] In the second state 200b, the overlapped area of the flexible display 240 and the first hinge structure 202 when the first hinge cover 250 is viewed from above may be wider than the overlapped area of the flexible display 240 and the second hinge structure 203 when the second hinge cover 260 is viewed from above. The first distance d1, which is the width of the first hinge cover 250 corresponding to the fourth display area 240d, may be longer (greater) than the second distance d2, which is the width of the second hinge cover 260 corresponding to the fifth display area 240e (e.g., the fifth display area 240e of FIG. 2A). The difference between the first distance d1 and the second distance d2 may correspond to the thickness of the second housing 220.

[0059] Based on the difference between the first distance d1 and the second distance d2, a preferentially foldable housing may be determined among the second housing 220 and the third housing 230. For example, the third housing 230 may preferentially rotate with respect to the first housing 210 based on the second folding shaft f2. After the third housing 230 is folded, the second housing 220 may rotate with respect to the first housing 210 based on the first folding shaft f1. Since the width of the second hinge structure 203 is narrower than the width of the first hinge structure 202, the third housing 230 may be folded to face the first housing 210. For example, since the width of the first hinge cover 250 is wider than that of the second hinge cover 260, the third housing 230 may be folded to face the first housing 210 based on the operation of the second hinge structure 203 in the second hinge cover 260. Since the width of the first hinge structure 202 is wider than the width of the second hinge structure 203, the second housing 220 may be folded to face the rear surface 231 of the third housing 230. For example, since the width of the first hinge cover 250 is wider than the width of the second hinge cover 260, the second housing 220 may be folded to face the rear surface 231 of the third housing 230 based on the operation of the first hinge structure 202 in the first hinge cover 250.

[0060] According to the above-described embodiment, the electronic device 200 may provide the flexible display 240 that may be folded at least once by including a plurality of hinge structures 202 and 203. The flexible display 240 may be folded several times, through the different width of hinge structures 202 and 203.

[0061] FIG. 3A illustrates an example of arrangement of a hinge locking structure in an unfolded state of an example electronic device according to various embodiments. FIG. 3B illustrates an example hinge locking structure in an unfolded state of an example electronic device according to various embodiments.

[0062] Referring to FIG. 3A, the electronic device 200 may be unfolded in the first state 200a. The flexible display 240 may be disposed on a planar surface formed by the first housing 210, the second housing 220, and the third housing 230.

[0063] The first hinge structure 202 may rotatably connect one side surface 221 of the second housing 220 facing the first side surface 211 of the first housing 210 to the first housing 210. The first angle θ_1 between the first housing 210 and the second housing 220 may be approximately 0 degrees to 180 degrees, through the rotation of the second housing 220. The second hinge structure 203 may rotatably connect one side surface 231 of the third housing 230 facing the second side surface 212 opposite to the first side surface 211 to the first housing 210. The second angle θ_2 between the first housing 210 and the third housing 230 may be approximately 0 degree to 180 degrees, through rotation of the third housing 230.

[0064] The first magnet 201 may be disposed in the second housing 220. The first magnet 201 may be disposed on a periphery portion 229 of the second housing 220 including another side surface 222 of the second housing 220 facing one side surface 221 of the second housing 220.

[0065] A hinge locking structure 330 may be disposed in the second hinge structure 203. According to an embodiment, the second hinge structure 203 may include the hinge locking structure 330. The second hinge structure 203 may further include a physical structure for the hinge locking structure 330.

[0066] Referring to FIG. 3B, the second hinge structure 203 may include at least one shaft 310 and a shaft holder 320. For example, the second hinge structure 203 may rotate the third housing 230 with respect to the first housing 210 based on the second folding shaft f2.

[0067] The at least one shaft 310 may rotate through a change in a folded state or an unfolded state of the electronic device 200. The shaft holder 320 may rotatably support at least one shaft 310. The at least one shaft 310 may rotate a gear (not illustrated). The shafts in the second hinge structure 203 may be interlocked by the gear.

[0068] The shaft holder 320 may include a first holder 320a supporting the side surface of the shaft 310 and a second holder 320b supporting the lower part of the shaft 310. The shaft holder 320 may include a space 321 for the hinge locking structure 330. The space 321 may guide the movement of the hinge locking structure 330. For example, the hinge locking structure 330 in the space 321 may move toward the shaft 310. The hinge locking structure 330 may move to the outside of the shaft holder 320 through an opening 322 facing the shaft 310. A stopper 323 of the shaft holder 320 may limit movement of the hinge locking structure 330 in the first direction (e.g., the z axis direction) opposite to the second direction (e.g., the -z axis direction) toward which the opening 322 is directed. For example, the seating space 321 may include the stopper 323 formed on an inner surface. The stopper 323 may include a stepped portion of the inner surface of the seating space 321. As the hinge locking structure 330 moves in the first direction (e.g., the z axis direction), the hinge locking structure 330 may contact the stopper 323. As the hinge locking structure 330 contacts the stopper 323, movement in the first direction (e.g., the z axis direction) may be limited.

[0069] The hinge locking structure 330 may include a pin 331 and a head 332. The pin 331 may be movable from the inside of the seating space 321 to the outside through the opening 322. The pin 331 may be formed in a cylindrical shape, but is not limited thereto. The pin 331 may have various cross-sections and may have a length in the first direction (e.g., the z axis direction) or the second direction

(e.g., the $-z$ axis direction). The head 332 may be in contact with one end of the pin 331. For example, the head 332 may be in contact with one end of the pin 331 facing the first direction (e.g., the z axis direction). For example, the head 332 may be in contact with one end facing the flexible display 240. For example, the pin 331 may be extended from one surface of the head 332 in the second direction (e.g., the $-z$ axis direction).

[0070] The hinge locking structure 330 may be in contact with the second magnet 340. The second magnet 340 may be attached to a portion of the hinge locking structure 330. The second magnet 340 may be disposed on one surface of the head 332 of the hinge locking structure 330. The second magnet 340 may face the flexible display 240. The second magnet 340 may interact with the first magnet 201. For example, in the first state 200a and the second state 200b, the distance between the first magnet 201 and the second magnet 340 may be far. Within a designated distance, the first magnet 201 and the second magnet 340 may move the hinge locking structure 330 in the second direction (e.g., the $-z$ axis direction) due to the repulsive force. For example, within the abnormal state of FIGS. 4A and 4B to be described later, the distance between the first magnet 201 and the second magnet 340 may be within a designated distance. Through the repulsive force of the first magnet 201 and the second magnet 340 disposed within the designated distance, the hinge locking structure 330 may move in the second direction (e.g., the $-z$ axis direction).

[0071] The electronic device 200 may further include an elastic member 350. The elastic member 350 may be disposed between the head 332 and the opening 322. The elastic member 350 may be in contact with the head 332. The elastic member 350 may be in contact with the inner surface of the seating space 321 in which the opening 322 is formed.

[0072] The elastic member 350 may elastically support the hinge locking structure 330 in the seating space 321. The elastic member 350 may be disposed within the seating space 321 to surround the pin 331 of the hinge locking structure 330. For example, the elastic member 350 may surround the outer circumferential surface of the pin 331. For example, the elastic member 350 may be a spring. However, the elastic member 350 is not limited thereto, and may include a rubber or porous material. For example, the elastic member 350 may be compressed by an external force. The elastic member 350 may be restored when the external force is removed.

[0073] The elastic member 350 may be compressed by the first magnet 201 and the second magnet 340 being within the designated distance. Through the interaction of the first magnet 201 and the second magnet 340 (e.g., repulsive force), the hinge locking structure 330 coupled to the second magnet 340 may move in a direction away from the first magnet 201 (e.g., $-z$ axis direction).

[0074] The shaft 310 may include a recessed groove 360 formed on a part of the outer circumferential surface of the shaft 310 and a protrusion 370 disposed in the groove 360. At least a part of the groove 360 may be overlapped with the hinge locking structure 330 along the outer circumferential surface of the shaft 310, when the second magnet 340 is viewed from above. For example, the hinge locking structure 330 may be disposed on the groove 360. The hinge locking structure 330 may be spaced apart from the groove 360 by an elastic force caused by the elastic member 350, or

be accommodated in at least a part of the space of the groove 360 through the interaction between the first magnet 201 and the second magnet 340.

[0075] The groove 360 may include stoppers 360a, 360b that limit the rotation of the shaft 310 while at least a part of the pin 331 is inserted into the groove 360. The stoppers 360a and 360b may be sidewalls of the groove 360. For example, the stoppers 360a and 360b may be sidewalls extending substantially vertically from one surface of the groove 360. The first stopper 360a and the second stopper 360b may be disposed in the periphery of the groove 360 formed by being recessed from the outer circumferential surface of the shaft 310 toward the one surface of the groove 360. The first stopper 360a may limit clockwise rotation of the shaft 310. For example, while the pin 331 is inserted into the groove 360, the pin 331 and the first stopper 360a may be contacted through clockwise rotation of the shaft 310. The shaft 310 cannot rotate clockwise due to the contact between the pin 331 and the first stopper 360a. The second stopper 360b may limit counterclockwise rotation of the shaft 310. For example, while the pin 331 is inserted into the groove 360, the pin 331 and the second stopper 360b may be contacted through counterclockwise rotation of the shaft 310. The shaft 310 cannot rotate counterclockwise due to the contact between the pin 331 and the second stopper 360b.

[0076] The groove 360 may include a plurality of protrusions 370. The plurality of protrusions 370 may protrude toward the outer circumferential surface of the shaft 310 within the groove 360. For example, the plurality of protrusions 370 may protrude from one surface of the groove 360 in a radial direction. One surface of the groove 360 may be a surface facing the radial direction of the shaft 310. The plurality of protrusions 370 may provide a plurality of accumulating spaces 361, 362, and 363 into which the pin 331 is insertable. The plurality of protrusions 370 may be spaced apart from each other by a designated interval. When the pin 331 is a polygon, the designated interval may correspond to the width of the pin 311. For example, the designated interval may correspond to a width in a rotation direction of the shaft 310 of the pin 311. When the pin 331 is circular, the designated interval may correspond to a diameter of an end surface of the pin 331. The designated interval may be related to the size of the accommodating space 362 between the plurality of protrusions 370 or the size of the accommodating spaces 361 and 363 between one of the protrusions and the sidewall of the groove 360. The designated interval may be an interval in which the pin 311 is insertable into the accommodating spaces 361, 362, and 363. For example, the designated interval may be equal to or larger than the width or diameter of the pin 311. Although the pin 311 has been described as being polygonal or circular, it is not limited thereto. The cross-section of the pin 311 may have an oval shape, or may have a shape, which is insertable into the groove 360 or the accommodating spaces 361, 362, and 363.

[0077] The plurality of protrusions 370 may include a first protrusion 370a and a second protrusion 370b. The plurality of protrusions 370 may provide a plurality of accommodating spaces 361, 362, and 363 into which the pin 331 is insertable. For example, the first protrusion 370a and the first stopper 360a may provide a first space 361. The first space 361 may be overlapped with the pin 331 when viewed in the z axis direction, since the third housing 230 rotates with respect to the first housing 210 in the first state 200a.

Since the rotation angle of the third housing 230 with respect to the first housing 210 is limited by providing the first stopper 360a, the first space 361 may prevent damage to the display in an abnormal state. The abnormal state will be described in detail below with respect to FIGS. 4A, 4B, 5A, and 5B.

[0078] The first protrusion 370a and the second protrusion 370b may provide a second space 362. The second space 362 may be overlapped with the pin 331, as the third housing 230 rotates with respect to the first housing 210 from the first state 200a. The second protrusion 370b and the second stopper 360b may provide the third space 363. The third space 363 may correspond to the position of the pin 331 in the first state 200a. The pin 331 may be accommodated in the third space 363 by an interaction between the first magnet 201 and the second magnet 340.

[0079] The plurality of protrusions 370 include the first protrusion 370a and the second protrusion 370b, and have been described as two protrusions, but this is only an example. The plurality of protrusions 370 may include one protrusion or may further include three or more protrusions, corresponding to the width of the groove 360. According to an embodiment, the groove 360 may not include a plurality of protrusions. For example, the width of the groove 360 may have a size corresponding to the diameter of the pin 331 or the width of the pin 331, and may not include protrusions. For example, in a case that the groove 360 has a width larger than the diameter of the pin 331 or the width of the pin 331, the shaft 310 may rotate between the first stopper 360a and the second stopper 360b of the groove 360.

[0080] The plurality of protrusions 370 and the side surface of the groove 360 may be parallel to the radial direction of the shaft. The side surface of the plurality of protrusions 370 may incline with respect to a radial direction of the shaft. The side surface having the inclinations of the plurality of protrusions 370 may allow a pin located in one of the spaces 361, 362, and 363 to move to another space. According to the movement of the pin 331, the plurality of protrusions 370 may provide physical or tactile notifications to the user of the electronic device 200. For example, the pin 331 of the hinge locking structure 330 may move along the side surface of the second protrusion 370b having an inclination from one (e.g., the third accommodating space 363) of the accommodating spaces 361, 362, and 363. The pin 331 may move to an accommodating space (e.g., the second accommodating space 362) next (adjacent) to the one accommodating space 363, through the movement in the z axis direction along the side surface of the second protrusion 370b.

[0081] FIG. 4A illustrates an example of arrangement of a hinge locking structure in a third state of an example electronic device according to various embodiments. FIG. 4B illustrates an example hinge locking structure in a third state of an example electronic device according to various embodiments. Referring to FIG. 4A, the first hinge structure 202 may rotatably connect one side surface 221 of the second housing 220 facing the first side surface 211 of the first housing 210 and the first housing 210. The second hinge structure 203 may rotatably connect one side surface 231 of the third housing 230 facing the second side surface 212 opposite to the first side surface 211 and the first housing 210.

[0082] In the third state 200c of the electronic device 200, the first housing 210 and the second housing 220 may be

folded, and the first housing 210 and the third housing 230 may be unfolded. The third state 200c may be a state in which the second housing 220 rotates with respect to the first housing 210 prior to rotation of the third housing 230 with respect to the first housing 210. Although the third housing 230 should preferentially rotate with respect to the first housing 210 during the folding operation of the electronic device 200, in the side surface where the second housing 220 preferentially rotates with respect to the first housing 210, the third state 200c may be referred to as an abnormal state. For example, in the third state 200c, an angle formed by the second housing 220 and the first housing 210 may be an angle- $\theta 1'$. The third state 200c may be a state in which the second housing 220 is further rotated by an angle $\theta 1'$ in the clockwise direction after the second housing 220 rotates 180 degrees clockwise in an unfolded state. In the abnormal state of the electronic device, since the third housing 230 cannot be folded to face the first housing 210, the electronic device may be damaged when pressed to switch to the folded state. The flexible display 240 may be disposed on a curved surface or a planar surface formed by the first housing 210, the second housing 220, and the third housing 230. For example, the flexible display 240 disposed on the first housing 210 and the second housing 220 may include a planar area and a curved area. The first display area 240a disposed on the first housing 210 and the second display area 240b disposed on the second housing 220 may be a planar area. The fourth display area 240d disposed between the first display area 240a and the second display area 240b may include a curved area. The fourth display area 240d may be bent according to folding of the first housing 210 and the second housing 220 by the first hinge structure 202. The first display area 240a, the third display area 240c, and the fifth display area 240e of the flexible display 240 disposed on the first housing 210 and the third housing 230 may be substantially form a planar surface. For example, since rotation of the third housing 220 according to the second hinge structure 203 does not exist, the first housing 210 and the third housing 230 may be identical to the first state 200a.

[0083] When converted from the first state 200a (e.g., the first state 200a in FIG. 2A) to the second state 200b (e.g., the second state 200b in FIG. 2B), the third housing 230 may be folded with respect to the first housing 210, before the second housing 220 is folded with respect to the first housing 210. The abnormal state of the electronic device may be a state in which the device is folded differently from the above-described folding order. For example, in the third state 200c, the second housing 220 may be folded to the first housing 210 before the third housing 230 is folded to the first housing 210. For example, the second display area 240b disposed on the second housing 220 may face the first display area 240a disposed on the first housing 210. The first magnet 201 adjacent to another side surface 222 of the second housing 220 may face the second hinge structure 203. The first magnet 201 may interact with the magnetic body in the second hinge structure 203. Within the third state 200c in which a part of the surface of the first housing 210 contacts the second surface 220a of the second housing 220, the hinge locking structure 330 in the second hinge structure 203 may face the first magnet 201. The operation of the hinge locking structure by the interaction between the first magnet 201 and the magnetic body (e.g., the second magnet 340) in the second hinge structure 203 will be described in FIG. 4B.

[0084] Referring to FIG. 4B, the electronic device 200 may further include a hinge locking structure 330. The hinge locking structure 330 may include the pin 331. In the third state 200c, the hinge locking structure 330 may be configured to move away from the first magnet 201 by the repulsive force between the first magnet 201 and the second magnet 340 within the seating space 321, and insert the pin 331 into the groove 360. The pin 331 may be inserted into the groove 360 formed in at least one shaft 310 within the second hinge structure 203. In the third state 200c, the pin 331 may protrude to the outside of the shaft holder 320 through the opening 322 by the interaction of the first magnet 201 and the second magnet 340 (e.g., repulsive force). In the third state 200c, the polarity of the first magnet 201 may be the same as the polarity of the second magnet 340. In the third state 200c, the first magnet 201 and the second magnet 340 may provide pushing force to each other by the interaction. For example, the first magnet 201 may provide a repulsive force (pushing the second magnet 340) to the second magnet 340. The second magnet 340 may provide a repulsive force pushing the first magnet 201 to the first magnet 201. Since the first magnet 201 is fixed to the inner space close to the other side surface 222 of the second housing 220, the hinge locking structure 330 may be moved from the inside of the seating space 321 to the outside by the force transmitted to the second magnet 340. For example, the hinge locking structure 330 may move the pin 331 to the inside of the seating space 321 through the second magnet 340 attached (or connected) to the head 332 of the hinge locking structure 330. The pin 331 protruding to the outside of the shaft holder 320 may occupy at least a part of the groove 360 of the shaft 310. For example, the pin 331 may be inserted into the third space 363 located under the hinge locking structure 330 in the third state 200c. The pin 331 inserted into the third space 363 of the groove 360 may limit rotation of the third housing 230 with respect to the first housing 210. In a state in which the pin 331 is inserted into the third space 363, rotation of the shaft 310 may be limited. When the shaft 310 rotates counterclockwise, the pin 331 and the second stopper 360b may contact each other, and thus rotation may be limited. When the shaft 310 rotates clockwise, the pin 331 and the second protrusion 370b may contact each other, and thus rotation may be limited. The pin 331 may ride over the second protrusion 370b when the second protrusion 370b has an inclined surface, but may give a physical notification to the user.

[0085] While the hinge locking structure 330 is switched from the third state 200c to the first state 200a, the pin 331 may move from the groove 360 to the seating space 321 by the elastic member 350. For example, when converted to the first state 200a, the distance between the first magnet 201 and the second magnet 340 may increase. As the repulsive force applied to the second magnet 340 is weakened, the elastic force of the elastic member 350 may be greater than the repulsive force. The elastic member 350 may push up the head 332. As the head 332 moves, the pin 331 integrally formed with the head 332 may move into the seating space 321. According to the movement of the head 332, the pin 331 integrally formed with the head 332 may move into the seating space 321.

[0086] According to the above-described embodiment, in the third state 200c, the electronic device 200 may limit the movement of the third housing 230 or may provide a notification to the user when the third housing 230 moves.

[0087] FIG. 5A illustrates an example of arrangement of a hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments. FIG. 5B illustrates an example hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments.

[0088] Referring to FIG. 5A, in the third state 200c (e.g., the third state 200c of FIG. 4A) of the electronic device 200, the third housing 230 may rotate with respect to the first housing 210 through the second hinge structure 203. The fourth state 200d may be one of the abnormal states of the electronic device. For example, the abnormal state may include the third state 200c in which the second housing 220 contacts a part of the first housing 210, a state of switching from the third state 200c to the fourth state, and a fourth state 200d in which a part of the third housing 230 contacts the side surface of the second housing 220 while the second housing 220 contacts a part of the first housing 210.

[0089] Even within the fourth state 200d, positions of the first housing 210 and the second housing 220 may be the same as or similar to those of the third state 200c.

[0090] In the electronic device 200 in the fourth state 200d, the first housing 210 and the second housing 220 may be folded, and the first housing 210 and the third housing 230 may be folded. The third housing 230 may be in contact with a part of the second housing 220. The fourth state 200d may be a state in which the third housing 230 rotates counterclockwise by a third angle θ_3 from the third state 200c. When the third housing 230 rotates counterclockwise further than the third angle θ_3 so that the third housing 230 faces the first housing 210, the flexible display 240 or the electronic device 200 may be damaged.

[0091] In the fourth state 200d, the first display area 240a and the second display area 240b of the flexible display 240 may face each other. The second housing 220 may contact the fifth display area 240e. In the fourth state 200d, the angle of the third display area 200c with the first display area 200a may be difficult to be further reduced. For example, the third housing 230 may be movable within the designated angular range to the first housing 210, within the third state 200c among the abnormal states.

[0092] The designated angular range may be up to approximately 180 degrees from the angle θ_2' between the first housing 210 and the third housing 230 in the fourth state 200d. For example, in the fourth state 200d, the angle θ_2' between the first housing 210 and the third housing 230 adjacent to the second housing 220 may be related to the position where the third housing 230 can no longer rotate counterclockwise. The angle at which the third housing 230 is moveable in the abnormal state may be from an angle θ_2' between the first surface 210a and the third surface 230a adjacent to the second housing 220 in the fourth state 200d in which the third surface 230a is adjacent to the second housing 220, to an angle θ_2 (e.g., approximately 180 degrees) between the first surface 210a and the third surface 230a, for example, in the first state 200a or the third state 200c.

[0093] The flexible display 240 may be disposed on a curved surface or a planar surface formed by the first housing 210, the second housing 220, and the third housing 230. For example, the flexible display 240 disposed on the first housing 210 and the second housing 220 may include a planar area and a curved area. The first display area 240a

disposed on the first housing **210** and the second display area **240b** disposed on the second housing **210** may be planar areas. The fourth display area **240d** disposed between the first display area **240a** and the second display area **240b** may include a curved area. The fourth display area **240d** may be bent by the first hinge structure **202**, according to the folding of the first housing **210** and the second housing **220**. The first display area **240a**, the third display area **240c**, and the fifth display area **240e** of the flexible display **240** disposed on the first housing **210** and the third housing **230** may include a planar area and a curved area. The first display area **240a** disposed on the first housing **210** and the third display area **240c** disposed on the third housing **230** may be a planar area. The fifth display area **240e** disposed between the first display area **240a** and the second display area **240b** may include a curved area. The fifth display area **240e** may be bent by the second hinge structure **203**, according to the folding of the first housing **210** and the third housing **230**.

[0094] The first magnet **201** may interact with a magnetic body (e.g., the second magnet **340**) in the second hinge structure **203**. In the third state **200c** in which a part of the first surface **210** of the first housing is adjacent to the second surface **220a** of the second housing (**220**), the hinge locking structure **330** within the second hinge structure **203** may face the first magnet **201**. The operation of the hinge locking structure **330** by the interaction of the magnetic body within the first magnet **201** and the second hinge structure **203** will be described in FIG. 5B.

[0095] Referring to FIG. 5B, the hinge locking structure **330** may operate within the influence of the reactive force acting between the first magnet **201** and the second magnet **340** even while switching from the third state **200c** to the fourth state **200d**. In the third state **200c**, when the third housing **230** moves, the pin **331** may move from one space of the accommodating spaces **361**, **362**, and **363** to another space in contact with the one protrusion, by moving along the surface of one of the plurality of protrusions **370**. In the fourth state **200d**, the pin **331** may be accommodated in the first space **361**. The pin **331** may be in contact with the first stopper **360a**. The hinge locking structure **330** may disengage the pin **331** of the hinge locking structure **330** from the groove **360** by the elastic member **350**, by deviating from the third state **200c** or the fourth state **200d**, which is the abnormal state. For example, the shaft **310** from which the pin **331** is removed may be switched to a first state **200a**, which is an unfolded state. The rotation of the shaft **310** in the clockwise direction may be limited by the pin **331** in contact with the first stopper **360a**. In the fourth state **200d**, while the second housing **220** rotates counterclockwise from the first housing **210**, the repulsive force acting between the first magnet **201** and the second magnet **340** may be changed in the hinge locking structure **330**. For example, the elastic member **350** may be changed based on a change in the repulsive force. The pin **331** may move from the groove **360** (e.g., the first space **361**) into the seating space **321** by the elastic member **350**.

[0096] According to the above-described embodiment, the hinge locking structure **330** may reduce damage to the flexible display **240** and the electronic device **200** within the fourth state **200d**, by limiting the counterclockwise rotation of the third housing **230**. For example, in the hinge locking structure **330**, as the elastic member **350** is compressed, the pin **331** may move outside the shaft holder **320**. The pin **331** moved to the outside of the shaft holder **320** may occupy one

of the grooves **360** or a plurality of spaces **361**, **362**, and **363** in the groove **360**. The hinge locking structure **330** may be configured to operate in an abnormal state (e.g., the third state **200c** or the fourth state **200d**) of the electronic device. The electronic device **200** may limit the rotation of the shaft **310** through the hinge locking structure **330** within the abnormal state. The electronic device **200** may prevent damage to the flexible display **240** and the electronic device **200** using the shaft **310** having limited rotation.

[0097] FIG. 6 is a block diagram of an example electronic device according to various embodiments.

[0098] Referring to FIG. 6, the electronic device **600** (e.g., the electronic device **101** of FIG. 1 or the electronic device **200** of FIG. 2A) may include a processor **620** (e.g., the processor **120** of FIG. 1), sensors **641** and **646** (e.g., the sensor module **176** of FIG. 1), and an output device **660** (e.g., the sound output module **155** of FIG. 1 or the display module **160**).

[0099] The electronic device **600** may control the first sensor **641**, the second sensor **646**, and the output device **660** through the processor **620**. The processor **620** may obtain data on the state of the electronic device **600** through the first sensor **641** and the second sensor **646**.

[0100] The processor **620** may be electrically connected to the first sensor **641**. The processor **620** may obtain first data related to the posture of the second housing (e.g., the second housing **220** of FIG. 2A) through the first sensor **641**. The first sensor **641** may be a sensor capable of obtaining sensing data related to the posture of the second housing **220**. For example, the first sensor **641** may obtain data related to an angle between the first housing **210** and the second housing **220**.

[0101] The processor **620** may be electrically connected to the second sensor **646**. The processor **620** may obtain second data related to the posture of the third housing (e.g., the third housing **230** of FIG. 2A) through the second sensor **646**. The second sensor **646** may be a sensor capable of obtaining sensing data related to the posture of the third housing **230**. For example, the second sensor **646** may obtain data related to an angle between the first housing **210** and the third housing **230**.

[0102] The first sensor **641** and the second sensor **646** may obtain data related to the distance, angle, posture sensed, and/or changes in magnetic field. Based on the distance, angle, posture, and/or the change in magnetic field, the processor **620** may identify an angle between the first housing **210** and the second housing **220** and/or an angle between the first housing **210** and the third housing **230**. For example, the first sensor **641** may be disposed in the first housing **210** and/or the second housing **220**. The first sensor **641** may include, for example, a proximity sensor, a gyro sensor, or a hall sensor. For example, when disposed in the first housing **210**, the first sensor **641**, which is a proximity sensor or a hall sensor, may obtain data related to a distance between the first housing **210** and the second housing **220**. Although the first sensor **641** is described as being disposed in the first housing **210**, the first sensor **641** may also be disposed in the second housing **220**. The processor **620** may identify an angle between the first housing **210** and the second housing **220** based on the data related to the distance. For example, the first sensor **641**, which is a gyro sensor or an acceleration sensor, may be disposed in the second housing **220** rotating based on the first housing **210**. The first sensor **641** may identify the posture of the second housing

220. Based on the identified posture of the second housing **220**, the processor **620** may identify an angle between the first housing **210** and the second housing **220**.

[0103] The second sensor **646** may include, for example, a proximity sensor, a gyro sensor, or a hall sensor. For example, when disposed in the first housing **210**, the second sensor **646**, which is a proximity sensor or a hall sensor, may obtain data related to the distance between the first housing **210** and the third housing **230**. Although the second sensor **646** is described as being disposed in the first housing **210**, the second sensor **646** may also be disposed in the third housing **230**. The processor **620** may identify an angle between the first housing **210** and the third housing **230** based on the data related to the distance. For example, the second sensor **646**, which is a gyro sensor or an acceleration sensor, may be disposed in the third housing **230** rotating based on the first housing **210**. The first sensor **641** may identify the posture of the third housing **230**. Based on the identified posture of the third housing **230**, the processor **620** may identify an angle between the first housing **210** and the third housing **230**.

[0104] Based on the first data obtained through the first sensor **641** and the second data obtained through the second sensor **646**, the processor **620** may identify the state of the first housing **220** or the third housing **230** based on the first housing **210**. For example, the processor **620** may identify the first angle θ_1 between the first housing **210** and the second housing **220** or the second angle θ_2 between the first housing **210** and the third housing **230**, based on the first data and the second data. For example, the processor **620** may identify that the electronic device **200** is in the first state **200a** based on identifying that the first angle θ_1 is approximately 180 degrees. The processor **620** may identify that the electronic device **200** is in the second state **200b** based on identifying that the first angle θ_1 is approximately 0 degrees. For example, the processor **620** may identify that the electronic device **200** is in the first state **200a** based on identifying that the second angle θ_2 is approximately 180 degrees. The processor **620** may identify that the electronic device **200** is in the second state **200b** based on identifying that the second angle θ_2 is approximately 0 degrees. According to the state change of the electronic device **200**, the first angle θ_1 and the second angle θ_2 may change. For example, as the electronic device **200** changes from the first state **200a** to the second state **200b**, the first angle θ_1 and the second angle θ_2 may change to approximately 0 degrees from 180 degrees.

[0105] The processor **620** may be configured to provide a first notification to the outside through output device **660**, based on the first angle θ_1 between the first housing **210** and the second housing **220** and the second angle θ_2 between the first housing **210** and the third housing **230**. The output device **660** may include a display module **160** or a sound output module **155**. The display module **160** may visually provide a notification based on the first angle θ_1 and the second angle θ_2 . The sound output module **155** may audibly provide a notification based on the first angle θ_1 and the second angle θ_2 .

[0106] The processor **620** may identify the abnormal state based on the first angle θ_1 being less than the second angle θ_2 . The abnormal state may be a state in which the postures of the first housing **210**, the second housing **220**, and the third housing **230** are different from previously designated postures. The designated postures may be postures in which the second housing **220** is foldable after the third housing

230. The first hinge structure (e.g., the first hinge structure **202** of FIG. 2A) rotatably connecting the second housing **220** to the first housing **210** may have a wider width than the second hinge structure (e.g., the second hinge structure **203** of FIG. 2A) rotatably connecting the third housing **230** to the first housing **210**. Due to the wide width of the first hinge structure **202**, the second housing **220** may be folded after the third housing **230** and changed to the second state **200b** (e.g., the second state **200b** of FIG. 2B). In order to be a fully folded second state (e.g., the second state **200b** of FIG. 2B), the third housing **230** may be folded on the first housing **210**, and the second housing **220** may be disposed on the folded third housing **230** (e.g., the +z axis direction). In a case that the electronic device **600** cannot be in a fully folded state (e.g., the second state **200b**) based on the first angle θ_1 and the second angle θ_2 , the processor **620** may provide a first notification to the outside through the output device **660**.

[0107] The processor **620** may provide another second notification distinguished from the first notification to the outside in response to identifying that the first angle θ_1 is less than or equal to 0 degrees. The processor **620** may identify that the first angle θ_1 is less than or equal to 0° while the first angle θ_1 is less than the second angle θ_2 . A case in which the first angle θ_1 is smaller than the second angle θ_2 within the designated angular range of the first angle θ_1 and the second angle θ_2 may be an abnormal state of the electronic device **600**. For example, in a case that the first angle θ_1 is smaller than the second angle θ_2 within the range of the first angle θ_1 and the second angle θ_2 from 0 degrees to 90 degrees, the state of the electronic device **600** may be an abnormal state. When the first angle θ_1 is 0 degrees or the first angle θ_1 is less than 0 degrees, the state of the electronic device **600** may be a third state **200c** (e.g., the third state **200c** of FIG. 4A). When the second angle θ_2 between the third housing **230** and the first housing **210** is less than the designated angle by operating the third housing **230**, while a state of the electronic device **600** is the third state **200c**, the flexible display (e.g., the flexible display **240** of FIG. 2A) or the electronic device **600** may be damaged. The processor **620** may provide a notification related to the possibility of damage to the flexible display **240** or the electronic device **600** to the user through the output device **660**. The processor **620** may identify that the second angle θ_2 between the third housing **230** and the first housing **210** is less than or equal to the designated angle. The processor **620** may be configured to provide a third notification to the outside based on the identification that the second angle θ_2 is less than or equal to the designated angle.

[0108] The electronic device **600** may identify an abnormal state of the electronic device by identifying the first angle θ_1 between the first housing **210** and the second housing **220** and the second angle θ_2 between the first housing **210** and the third housing **230**. The processor **620** may provide a different notification to the user according to the degree of the abnormal state based on the identified abnormal state. The user may stop the folding operation by recognizing the possibility of damage to the display or the electronic device through the provided notification.

[0109] Considering that the flexible display **240** is covered by the housings **210**, **220**, and **230** in the second state **200b**, the electronic device **200** may be referred to as an in-folding electronic device. The electronic device **200** of the above-described embodiment has been described based on the in-folding electronic device, but is not limited thereto. The

electronic device may be an out-folding electronic device in which the flexible display **240** is disposed on the outer surface of the electronic device and exposed in the folded state. For example, the first angle $\theta 1$ and the second angle $\theta 2$ described above may vary within a range of 180 degrees to 360 degrees. Referring back to FIG. 2A, in the unfolded state, the second housing **220** may rotate counterclockwise based on the first housing **210**, and the third housing **230** may rotate clockwise based on the first housing **210**.

[0110] The second housing **220** may be folded with respect to the first housing **210**, based on the hinge structure or the width of the hinge cover being different, after the third housing **230** is folded with respect to the first housing **210** similar to the above-described embodiment. In order to prevent damage when the second housing **220** is preferentially folded, the third magnet (not illustrated) and the fourth magnet (not illustrated) may be disposed at the same/similar positions to the arrangement positions of the first magnet **201** and the second magnet **340**.

[0111] FIG. 7 is a flowchart illustrating an example operation of detecting an abnormal state of an electronic device according to various embodiments.

[0112] Referring to FIG. 7, in operation **710**, the processor **620** (e.g., the processor **620** of FIG. 6) may obtain the first angle $\theta 1$ and the second angle $\theta 2$ based on the first data and the second data sensed through the first sensor (e.g., the first sensor **641** in FIG. 6) and the second sensor (e.g., the second sensor **646** in FIG. 6). The first data may be a distance or an angle between the first housing **210** and the second housing **220** obtained through the first sensor **641**. The second data may be a distance or an angle between the first housing **210** and the third housing **230** obtained through the second sensor **646**.

[0113] In operation **720**, the processor **620** may provide a notification to the outside based on the first angle $\theta 1$ and the second angle $\theta 2$. The processor **620** may identify the state of the electronic device based on the first angle $\theta 1$ and the second angle $\theta 2$. The processor **620** may provide various notifications to the outside based on the state of the electronic device. For example, in response to identifying that the state of the electronic device is the abnormal state, a first notification related to the abnormal state of the electronic device may be provided externally. For example, in a case that the state of the electronic device is identified as the third state (e.g., the third state **200c** of FIG. 4A), the processor **620** may provide a second notification to warn that the display or electronic device may be damaged when the third housing is moved more than the designated angle. For example, in a case that the state of the electronic device is identified as the fourth state (e.g., the fourth state **200d** of FIG. 4A), the processor **620** may provide a third notification to warn that the display or electronic device may be damaged when the third housing is further moved.

[0114] In FIGS. 6 and 7, the electronic device may be configured to prevent damage to the electronic device or the display by notifying the abnormal state of the electronic device by identifying the first angle $\theta 1$ and the second angle $\theta 2$. The electronic device may physically limit the movement of the third housing **230** in the abnormal state (e.g., the third state **200c** or the fourth state **200d**) by having a hinge structure including the hinge locking structure **330** of FIGS. 3A to 5B.

[0115] According to the above-described example embodiments, the electronic device may reduce the possi-

bility of damage to the electronic device due to movement of the third housing, by providing a locking structure that physically limits the movement of the third housing when the abnormal state is reached. The electronic device may notify the wrong (potentially damaging) movement of the third housing, by alerting the user by providing a step-by-step notification of abnormal state.

[0116] According to the above-described example embodiments, an electronic device (e.g., the electronic device **101** of FIG. 1 or the electronic device **200** of FIG. 2A) may include a first housing (e.g., the first housing **210** of FIG. 2A) including a first surface (e.g., the first surface **210a** of FIG. 2A), a second housing (e.g., the second housing **220** of FIG. 2A) including a second surface (e.g., the second surface **220a** of FIG. 2A), and a third housing (e.g., the third housing **230** of FIG. 2A) including a third surface (e.g., the third surface **230a** of FIG. 2A).

[0117] According to an example embodiment, the electronic device may include a first hinge structure (e.g., the first hinge structure **202** of FIG. 2A) and a second hinge structure (e.g., the second hinge structure **260** of FIG. 2A). The first hinge structure may rotatably connect one side surface (e.g., one side surface **221** of FIG. 3A) of the second housing facing a first side surface (e.g., the first side surface **211** of FIG. 3A) of the first housing to the first housing. The second hinge structure may rotatably connect one side surface (e.g., one side surface **231** of FIG. 3A) of the third housing facing a second side surface (e.g., the second side surface **212** of FIG. 3A) of the first housing opposite to the first side surface to the first housing. The electronic device may further include a flexible display (e.g., the flexible display **240** of FIG. 2A). The flexible display may be disposed on a surface formed by the first housing, the second housing, and the third housing. According to an example embodiment, the electronic device may further include a first magnet (e.g., the first magnet **201** of FIG. 3A). The first magnet may be disposed in the second housing. According to an example embodiment, the electronic device may further include a hinge locking structure (e.g., the hinge locking structure **330** of FIG. 3B). The hinge locking structure may include a pin (e.g., the pin **331** of FIG. 3B) facing the first magnet and insertable into a groove formed in at least one shaft in the second hinge structure (e.g., the groove **360** of FIG. 3B), in an abnormal state of the electronic device in which a part of the first surface is adjacent to the second surface.

[0118] According to an example embodiment, the first hinge structure and the second hinge structure may switch the electronic device into an unfolded state in which the first surface, the second surface, and the third surface face the same direction, or a folded state in which at least one of the first surface, the second surface, and the third surface face different directions.

[0119] According to an example embodiment, the pin may be spaced apart (or disengaged) from the groove in the hinge locking structure as the abnormal state is changed to the unfolded state.

[0120] According to an example embodiment, the first magnet may be disposed in a periphery portion (e.g., the periphery portion **229** of FIG. 3A) of the second housing including the other side surface of the second housing facing the one side surface of the second housing.

[0121] According to an example embodiment, the hinge locking structure may include a second magnet (e.g., the

second magnet **340** of FIG. 3B). The second magnet may be disposed on a surface of the hinge structure facing the flexible display.

[0122] According to an example embodiment, the polarity of the first magnet may be the same as that of the second magnet.

[0123] According to an example embodiment, in the abnormal state of the electronic device, the hinge locking structure may be configured to insert the pin into the groove by moving from the first magnet within the seating space by the repulsive force between the first magnet and the second magnet.

[0124] According to an example embodiment, the second hinge structure may further include a shaft holder (e.g., the shaft holder **320** of FIG. 3B). The shaft holder **320** may rotatably support the at least one shaft.

[0125] According to an example embodiment, the shaft holder may include a seating space in which the hinge locking structure is seated. The shaft holder may include an elastic member (e.g., the elastic member **350** of FIG. 3B). The elastic member may elastically support the hinge locking structure within the seating space.

[0126] According to an example embodiment, while the electronic device is changed to the unfolded state from the abnormal state, the pin may be moved from the groove to the seating space by the elastic member.

[0127] According to an example embodiment, the at least one shaft may rotate as the electronic device changes to a folded state or an unfolded state, and may be configured to rotate the third housing with respect to the first housing.

[0128] According to an example embodiment, the groove may include a plurality of protrusions (e.g., the plurality of protrusions **370** of FIG. 3B) protruding in a radial direction of the shaft and spaced apart by a designated interval in the groove.

[0129] According to an example embodiment, the designated interval may correspond to the width (or other dimension) of the pin.

[0130] According to an example embodiment, the plurality of protrusions may provide a plurality of accommodating spaces into which the pin is insertable.

[0131] According to an example embodiment, a side surface of the groove may be parallel to the radial direction of the shaft.

[0132] According to an example embodiment, the side surface of the plurality of protrusions may have an inclination relative to a radial direction of the shaft.

[0133] According to an example embodiment, when the third housing moves within the abnormal state of the electronic device, the pin may move from one space (e.g., a third space **363**) of the accommodating spaces (e.g., the first space **361**, the second space **362**, and the third space **363**) to another space (e.g., a second space **362**) contacting the one protrusion by moving along the surface of one of the plurality of protrusions.

[0134] According to an example embodiment, the third housing may be movable within a designated angular range in the abnormal state of the electronic device with respect to the first housing.

[0135] According to an example embodiment, the designated angular range may be 180 degrees from an angle between the first surface contacting the second housing in the abnormal state and the third surface contacting the side surface of the second housing.

[0136] According to an example embodiment, in the unfolded state, an overlapped area of the display and the first hinge structure when the first hinge structure is viewed from above may be wider than an overlapped area of the display and the second hinge structure when the second hinge structure is viewed from above. For example, the width of the first hinge cover may be wider than the width of the second hinge cover.

[0137] According to an example embodiment, the electronic device may include a processor (e.g., the processor **120** of FIG. 1 or the processor **620** of FIG. 6), a first sensor (e.g., the first sensor **641** of FIG. 6), and a second sensor (e.g., the second sensor **646** of FIG. 6).

[0138] According to an example embodiment, the first sensor may obtain first data related to the posture of the second housing and may be disposed in the second housing.

[0139] According to an example embodiment, the second sensor may obtain second data related to the posture of the third housing and may be disposed in the third housing.

[0140] According to an example embodiment, the processor may obtain a positional relationship of the second housing or the third housing based on the first housing based on the first data and the second data sensed through the first sensor. The positional relationship may be expressed by an angle.

[0141] According to an example embodiment, the processor may be configured to provide a notification to the outside, in response to identifying that the first angle $\theta 1$ between the first housing and the second housing is less than the second angle $\theta 2$ between the first housing and the third housing.

[0142] According to an example embodiment, the processor may be configured to provide another notification distinguished from the notification to the outside in response to identifying that the first angle $\theta 1$ is 0 degrees.

[0143] According to an example embodiment, the pin inserted into the groove may limit rotation of the third housing with respect to the first housing.

[0144] According to an example embodiment, the electronic device (e.g., the electronic device **101** of FIG. 1 or the electronic device **200** of FIG. 2A) may include a first housing (e.g., the first housing **210** of FIG. 2A) including a first surface (e.g., the first surface **210a** of FIG. 2A), a second housing (e.g., the second housing **220** of FIG. 2A) including a second surface (e.g., the second surface **220a** of FIG. 2A), and a third housing (e.g., the third housing **230** of FIG. 2A) including a third surface (e.g., the third surface **230a** of FIG. 2A).

[0145] According to an example embodiment, the electronic device may include a first hinge structure (e.g., the first hinge structure **202** of FIG. 2A) and a second hinge structure (e.g., the second hinge structure **203** of FIG. 2A). The first hinge structure may rotatably connect one side surface (e.g., one side surface **221** of FIG. 3A) of the second housing facing a first side surface (e.g., the first side surface **211** of FIG. 3A) of the first housing to the first housing. The second hinge structure may rotatably connect one side surface (e.g., the side surface **231** of FIG. 3A) of the third housing facing a second side surface (e.g., the second side surface **212** of FIG. 3A) of the first housing opposite to the first side surface.

[0146] The example electronic device may further include a flexible display (e.g., the flexible display **240** of FIG. 2A).

The flexible display may be disposed on a surface formed by the first housing, the second housing, and the third housing.

[0147] According to an example embodiment, the electronic device may include a processor (e.g., the processor 120 of FIG. 1 or the processor 620 of FIG. 6), a first sensor (e.g., the first sensor 641 of FIG. 6), and a second sensor (e.g., the second sensor 646 of FIG. 6). According to an example embodiment, the first sensor may obtain first data related to the posture of the second housing and may be disposed in the second housing. According to an example embodiment, the second sensor may obtain second data related to the posture of the third housing and may be disposed in the third housing.

[0148] According to an example embodiment, the electronic device may include a first magnet within the second housing.

[0149] According to an example embodiment, the electronic device may include a hinge locking structure (e.g., the hinge locking structure 330 of FIG. 3B). The hinge locking structure may include a pin (e.g., the pin 331 of FIG. 3B) facing the first magnet and insertable into a groove formed in at least one shaft in the second hinge structure, in abnormal state of the electronic device in which a part of the first surface is adjacent to the second surface.

[0150] According to an example embodiment, the first hinge structure and the second hinge structure may be configured to provide an unfolded state in which the first surface, the second surface, and the third surface face the same direction, or a folded state in which at least one of the first surface, the second surface, or the third surface face different directions.

[0151] According to an example embodiment, an overlapped area between the display and the first hinge structure when the display is viewed from above in the unfolded state may be larger than an overlapped area between the display and the second hinge structure.

[0152] According to an example embodiment, the electronic device may further include a processor (e.g., the processor 620 of FIG. 6).

[0153] According to an example embodiment, the processor may obtain the angle of the second housing or the third housing based on the first housing based on the first data and the second data obtained through the first sensor, and provide a first notification to the outside in response to identifying that the first angle θ_1 between the first housing and the second housing is smaller than the second angle θ_2 between the first housing and the third housing.

[0154] According to an example embodiment, the processor may be configured to provide a second notification distinguished from the first notification to the outside in response to identifying that the first angle θ_1 is 0 degrees.

[0155] According to an embodiment, the processor may be configured to identify that the second angle θ_2 is less than or equal to the designated angle and provide a third notification to the outside based on an identification that the second angle θ_2 is below the designated angle.

[0156] An example electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance,

or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0157] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and do not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0158] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0159] While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those of ordinary skill in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

What is claimed is:

1. A foldable electronic device comprising:
 - a first housing part;
 - a second housing part;

a third housing part;
 a first hinge rotatably coupling the first housing part and the second housing part;
 a second hinge rotatably coupling the first housing part and the third housing part;
 a flexible display disposed on a first surface of the first housing part, a second surface of the second housing part and a third surface of the third housing part;
 at least one sensor;
 at least one processor comprising processing circuitry; and
 memory comprising one or more storage media storing instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to:
 using the at least one sensor, identify a folding state of the foldable electronic device, and
 based at least in part on the folding state of the foldable electronic device corresponding to an abnormal folded state, display a warning message related to the folding state via at least a portion of the flexible display.

2. The foldable electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to, based on an angle between the first housing part and the second housing part while identifying a state in which the third housing part is folded with respect to the first housing part, identify the abnormal folded state.

3. The foldable electronic device of claim 1, wherein the at least one sensor includes a first sensor and a second sensor, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to:
 using the first sensor, obtain a first data related to a first angle, a first distance, or first magnetic field between the first housing and the second housing,
 using the second sensor, obtain a second data related to a second angle, a second distance, or second magnetic field between the first housing and the third housing, and
 based at least in part on the first data and the second data, identify the folding state of the foldable electronic device.

4. The foldable electronic device of claim 1, wherein instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to display the warning message related to the folding state when the foldable electronic device is in the abnormal folded state.

5. The foldable electronic device of claim 1, wherein the warning message includes an advice to a user of the foldable electronic device regarding a future folding state of the foldable device.

6. The foldable electronic device of claim 1, wherein instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to display the warning message related to the folding state based on a determination that the foldable electronic device may be damaged if the third housing is moved more than a designated angle.

7. The foldable electronic device of claim 1, wherein instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to display the warning message related to the folding state based on a determination that the foldable electronic device may be damaged if the third housing is further moved.

8. The foldable electronic device of claim 1, wherein the at least one sensor includes at least one of a proximity sensor, a gyro sensor, a hall sensor or an acceleration sensor.

9. The foldable electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to:
 while the first housing part and the second housing part are unfolded with respect to each other, identify, via the at least one sensor, a folding movement of the first and third housing parts to the abnormal folded state, and
 based on identifying the first and third housing parts being moved to the abnormal folded state, display, via the flexible display, a notification regarding the abnormal folded state of the foldable electronic device.

10. The foldable electronic device of claim 9, wherein the abnormal folded state is defined within a rotational range of the third housing part being rotated with respect to the first housing part.

11. The foldable electronic device of claim 9, wherein the folding movement of the first and third housing parts to the abnormal folded state is identified via the at least one sensor while the third housing part is rotated with respect to the first housing part within a rotational range allowed by the second hinge.

12. The foldable electronic device of claim 9, wherein the abnormal folded state is defined within a rotational range from a first state in which the first and third housing parts are unfolded with respect to each other to a second state in which the first and third housing parts are folded with respect to each other.

13. The foldable electronic device of claim 9, wherein the abnormal folded state of the foldable electronic device is defined while allowing the third housing part to rotate with respect to the first housing part via the second hinge assembly.

14. The foldable electronic device of claim 1, wherein the at least one sensor includes:
 at least one first sensor for identifying a folding movement of the first and third housing parts, and
 at least one second sensor for identifying a folding movement of the first and second housing parts,
 wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to:
 identify an angle between the first housing part and the second housing part via the at least one second sensor,
 based on identifying that the angle between the first housing part and the second housing part is within a threshold angle range via the at least one second sensor, identify that the first housing part and the second housing part are folded, and

based on identifying that the angle between the first housing part and the second housing part is out of the threshold angle range via the at least one second sensor, identify that the first housing part and the third housing part are unfolded.

15. The foldable electronic device of claim **14**,

wherein the threshold angle range is a first threshold angle range, and

wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to identify the third housing part at least partially folded with respect to the first housing part by identifying that an angle between the first housing part and the third housing part is within a second threshold angle range via the at least one first sensor.

16. The foldable electronic device of claim **1**,

wherein the at least one sensor includes:

at least one first sensor for identifying a folding movement of the first and third housing parts, and

at least one second sensor for identifying a folding movement of the first and second housing parts,

wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to, based on identifying that the first housing part and the second housing part are unfolded via the at least one second sensor, enable identifying whether the third housing part is at least partially folded with respect to the first housing part via the at least one first sensor.

17. The foldable electronic device of claim **16**,

wherein identifying whether the third housing part is at least partially folded with respect to the first housing part via the at least one first sensor is disabled based on identifying that the first housing part and the second housing part are folded via the at least one second sensor.

18. The foldable electronic device of claim **1**,

wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to, based on identifying that the third housing part is at least partially folded with respect to the first housing part, enable identifying whether the first housing part and the second housing part are unfolded or folded, and

wherein identifying whether the first housing part and the second housing part are unfolded or folded is disabled based on identifying that the third housing part is not at least partially folded with respect to the first housing part.

19. The foldable electronic device of claim **1**,

wherein the second housing part is, when the first housing part and the second housing part are folded, rotated in a first rotational direction with respect to the first housing part, and

wherein the third housing part is, when the first housing part and the third housing part are folded, rotated in a second rotational direction opposite to the first rotational direction with respect to the first housing part.

20. The foldable electronic device of claim **1**,

wherein a width of the first hinge is different from a width of the second hinge.

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